DOCUMENTATION GAP REGISTER

Purpose. This register lists every item that the chapter content could not establish from the repository, every item that requires the author's input, every additional test that would need to be run for a claim to stand, and every documentation inconsistency that must be resolved before submission.

Prepared from: `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`, read-only inspection of source, migrations, tests, evaluation artefacts and project records.

How to use it. Each row names what is missing, why it matters, which chapter section depends on it, and what the author must supply. Rows marked **BLOCKING** must be closed before submission; **IMPORTANT** rows affect the strength of a claim but do not invalidate the chapter; **OPTIONAL** rows improve the report if closed.

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1. Items requiring author input (not obtainable from the repository)

ID	Item	Why it matters	Where used	What the author must supply	Priority
G-01	Workstation and device specification used for development and testing (processor, RAM, storage, OS build, browser and version, phone model if used)	Chapter 5 §5.2.1 states that hardware was not recorded. The run records show 7.4 GB RAM with ~0.5 GB free, but not the full specification	Ch.5 §5.2.1, §5.4.2	A specification sheet for the machines and devices actually used	**BLOCKING**
G-02	Confirmation of free memory during the sweeps that reached 149–150/151	Chapter 5 §5.4.3 reports the instability and identifies memory pressure as the mechanism, but the hardware cause is inferred rather than measured	Ch.5 §5.4.3	Free-memory observation at the time, or a clean re-run on a suitably provisioned machine	IMPORTANT
G-03	Statement of whether Figures 4.1–4.24 were produced, and their file names	The chapters specify each figure precisely but contain no drawings; the Word document needs rendered figures, not placeholders	Ch.4 §4.3, §4.5, §4.7	The drawn figures, or a decision to draw them from the specifications	**BLOCKING**
G-04	Screenshots for Figures 4.13–4.23 (sign-in, registry list, eligibility, registration form, screening confirmation, dashboard, duplicate review, awards and certificates, administrative support screens, both public verification pages)	The specifications state what each image must show; no screenshot was captured in this pass and none is claimed	Ch.4 §4.5; Ch.5 §5.2	Redacted screenshots taken against synthetic data only	**BLOCKING**
G-05	Confirmation that screenshots and quoted identifiers contain no real beneficiary data	The repository holds only synthetic data, but the live deployment holds demonstration records created during testing	Ch.5 §5.6; G-13	Visual check of every image before insertion	**BLOCKING**
G-06	User acceptance study: participants, roles, tasks, findings, consent and ethics approval — or an explicit statement that none was conducted	Chapter 5 §5.5.6 reports that UAT was not conducted with users; NFR-14 is unverified without it	Ch.5 §5.5.6; Ch.6 §6.4.9	Either the study record or an explicit statement of its absence	**BLOCKING**
G-07	Decision on how `docs/uat-test-plan.md` is treated: author walkthrough, marked superseded, or withdrawn	Its five `PASS` results and one stale sentence ("current backend controller code does not call it") conflict with the implementation and the test report	Ch.5 §5.5.6, §5.6.7	An instruction, since changing it is a documentation edit the author controls	IMPORTANT
G-08	Decision on how `docs/verification.md` is treated, given its claim that no controller calls the predictor	That claim is false for the current revision and would contradict Chapters 4 and 5 if quoted	Ch.5 §5.6.7	Annotation, retirement, or acceptance of the correction recorded here	IMPORTANT

ID	Item	Why it matters	Where used	What the author must supply	Priority
G-09	AI-assistance disclosure required by the School (tools used, how outputs were verified)	Many institutions require such a statement; none exists in the repository	Front matter / declaration	The wording the School requires	**BLOCKING**
G-10	Student name, admission number and supervisor name exactly as they must appear	Chapters 4–6 do not name the author and must agree with the title page and Chapters 1–3	Title page	Confirmation of the exact form (the proposal reads "Ngigi Peter", 169613, supervisor Salome Monthe Chemiat)	IMPORTANT

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2. Items requiring additional testing or evidence

ID	Item	Why it matters	Where used	What would close it	Priority
G-11	Phone-camera scan of a printed student QR code and of a printed award certificate	The identification mechanism is a printed code read by a phone; only headless-Chrome checks exist	Ch.5 §5.4.8, §5.6.3; Ch.6 §6.4.6	A camera test against the deployed HTTPS origin, with observed results recorded	**BLOCKING** for any claim that the mechanism works in practice
G-12	Load and concurrency evidence at 500 and 1 000 candidates, and two simultaneous registrations	NFR-01 is verified only at 50 candidates and NFR-13 has no concurrency test	Ch.5 §5.4.5, §5.5.4; Ch.6 §6.4.8	A scripted load test plus a concurrency test, with results recorded	IMPORTANT
G-13	Removal of the two `E2E Probe` students and their duplicate case from the production database	They are demonstration artefacts in a live database and would appear in any live screenshot	Ch.5 §5.3.12	Execute the documented `DELETE` for `full_name LIKE 'E2E Probe%'` (author's decision; not performed here)	IMPORTANT
G-14	Administrator password re-alignment and re-run of `scripts/admin-reveal-smoke.mjs`	The stored password returned 401, so the reveal screen was not re-checked after the registration-field change	Ch.5 §5.3.12	Reset the password, re-run the harness, record the result	IMPORTANT
G-15	Print-output verification of the certificate PDF and the exported report	Only the generated bytes are covered; no physical print or visual check exists	Ch.5 §5.4.8	Print both documents and record legibility and layout	OPTIONAL
G-16	Accessibility audit and cross-browser testing (Firefox, Safari, mobile)	NFR-14 and NFR-19 are unverified beyond Chrome	Ch.5 §5.4.8, §5.5.4	An accessibility tool report plus manual checks, and a cross-browser pass	OPTIONAL
G-17	Tests for the eligibility arithmetic, dashboard counter arithmetic and report content	Three implemented areas have weak automated coverage	Ch.5 §5.4.8 (O-07, O-08)	Table-driven tests added to the Jest suite	IMPORTANT
G-18	Adjudicated pair labels and a re-run of the reviewed-pairs evaluation	Every real-world quality claim depends on it; decisions D1–D5 are unanswered	Ch.5 §5.5.5; Ch.6 §6.4.1	200–300 adjudicated pairs plus the evaluation output	**BLOCKING** for a real-world accuracy claim only (not required while the report confines itself to synthetic findings, as these chapters do)
G-19	Hosted disaster-recovery drill	The rehearsal used a local disposable database, not the managed server	Ch.5 §5.4.5; Ch.6 §6.4.8	A restore exercise against a separate managed instance, with results recorded	OPTIONAL
G-20	One clean 151/151 suite run on a machine with adequate memory	The documentation-session sweeps reached 149–150/151 for an environmental reason; the report should show a clean reproduction	Ch.5 §5.4.3	One green sweep with the memory note recorded	IMPORTANT

3. Documentation inconsistencies to reconcile

ID	Inconsistency	Evidence	Affects	Suggested resolution
G-21	The proposal specifies a **Random Forest** classifier over **four** features; the production system runs **XGBoost** over **six**	Proposal §1.3.1(ii), §3.2.2, §3.2.3 versus `predict.py`, `feature_engineering.py`	Ch.4 §4.1, §4.4.4; Ch.5 §5.3.4; Ch.6 §6.2.4	Report the deviation explicitly (as these chapters do), and state a reason if one exists; otherwise report it as an unexplained deviation
G-22	The proposal describes deployment on "a single Linux virtual machine" with Nginx and PM2; the system runs on Azure Container Apps plus App Service	Proposal §3.3.7, §3.4.5 versus `docs/azure-deployment-log-2026-10-31.md`	Ch.4 §4.7; Ch.5 §5.3.12	Report the deployed architecture and note the change, with the hosting constraints that forced it
G-23	`docs/verification.md` states there are no callers of the prediction function and that controllers use their own scoring	`students.controller.js:69` calls `scorePairs`	Ch.5 §5.6.7	Mark the document superseded (see G-08)
G-24	`docs/uat-test-plan.md` reports five `PASS` results without participants and contains a stale sentence about the predictor	`docs/uat-test-plan.md`, Test Scenario 2	Ch.5 §5.5.6	Reclassify as an author walkthrough, or run a real study (G-06, G-07)
G-25	`docs/FINAL_TEST_REPORT.md` contains two result generations (21 suites/128 tests on 2026-10-31; 24 suites/151 tests on 2026-11-01)	§1–§3 versus §10	Ch.5 §5.4.3	Always cite the date with a count (as these chapters do); optionally add a "current totals" banner to the file
G-26	Four working checkouts of the same project exist on different branches, and their documents differ	`student-beneficiary-system`, `-dev`, `-admin-login-fix`, `sbms-raw-pair-model-training`	Evidence report	State which checkout and commit every claim refers to (these chapters use `fix/sbms-completion-work` @ `d6b5032`)
G-27	The production model artefact is tracked, but `.gitignore` still lists `ml-service/model/*.joblib`	`.gitignore` versus `git ls-files ml-service/model`	Ch.5 §5.3.4	Note that the artefact is committed; if it is meant to be excluded, explain how the deployment obtains it
G-28	Two root reports cited by these chapters are untracked (`SBMS_O10_O11_IMPLEMENTATION_REPORT.md`, `SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md`)	`git status` shows both as untracked	Ch.4 §4.4.3; Ch.5 §5.4.7 (D-08)	Commit them, or cite the committed equivalent

4. Items deliberately not supplied, and why

Item	Reason
Invented supervisor feedback or approval	The SBMS-01…SBMS-18 findings are unrecoverable from the workspace; `SBMS_PHASE_2_SUPERVISOR_FINDINGS_REVIEW.md` §4 records that their wording is unavailable. No finding was paraphrased into a requirement
Production metrics, latency under load, uptime figures	No such measurements exist
Real-world ML precision, recall or F1	No adjudicated labels exist; synthetic figures are reported as synthetic
User-acceptance results	No study was conducted
New academic citations	Chapters 4–6 introduce no new literature claims; where a concept needs a source, the citation belongs in Chapters 1–3. Any citation added during editing must be verified by the author
Authentication of the deployment as "production"	The deployment is a demonstration instance, and the chapters say so

5. Cross-reference: gaps to chapter sections

Chapter section	Gaps that touch it
Ch.4 §4.1	G-21
Ch.4 §4.2.3	G-12, G-16
Ch.4 §4.3	G-03
Ch.4 §4.5	G-03, G-04, G-05
Ch.4 §4.7	G-22
Ch.5 §5.2.1	G-01, G-02
Ch.5 §5.3.4	G-21, G-27
Ch.5 §5.3.12	G-13, G-14, G-22
Ch.5 §5.4.3	G-02, G-20, G-25
Ch.5 §5.4.5	G-12, G-19
Ch.5 §5.4.8	G-11, G-15, G-16, G-17
Ch.5 §5.5.5	G-18
Ch.5 §5.5.6	G-06, G-07
Ch.5 §5.6.7	G-08, G-23, G-24, G-25
Ch.6 §6.2.4	G-21
Ch.6 §6.4.1–§6.4.9	G-11, G-12, G-17, G-18, G-19

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6. Minimum set to close before submission

If time is short, five items carry the most risk: **G-01** (environment specification), **G-03/G-04/G-05** (figures and screenshots, with the synthetic-data check), **G-06/G-07** (a clear position on user acceptance), **G-21** (the Random Forest to XGBoost deviation) and **G-11** (a phone-camera scan of at least one printed QR code, since the identification mechanism otherwise rests entirely on a headless-browser check).

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7. Closed since this register was drafted

Item	How it was closed	Where the closure is visible
G-21 Random Forest to XGBoost deviation	Stated explicitly, not reconciled silently: the proposal's four-feature Random Forest and the delivered six-feature XGBoost are both named, with the research-only artifact identified	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` §4.1, §4.2.2 (FR-31), §4.4.4; `CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORKS.md` §6.2.4; restated in `CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md` §1

Still open, and unchanged by the above: G-21 remains an item the author must keep visible through final assembly, because a supervisor comparing Chapter Three to Chapter Four will meet the change there. The register keeps it listed for that reason rather than marking it "done".

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8. Document set this register belongs to

The evidence package is two-tiered. Entry set: `ACADEMIC_GUIDE_REQUIREMENTS.md`, `CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN_EVIDENCE.md`, `CHAPTER_5_IMPLEMENTATION_TESTING_AND_EVALUATION_EVIDENCE.md`, `CHAPTER_6_CONCLUSIONS_RECOMMENDATIONS_AND_FUTURE_WORK_EVIDENCE.md`, `CROSS_CHAPTER_CONSISTENCY_AND_TRACEABILITY.md`, `DIAGRAM_SPECIFICATIONS.md`, `MASTER_EVIDENCE_REGISTER.md`, `ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md`, `CLAUDE_HANDOFF_GUIDE.md`. Evidence tier: the three chapter drafts, the `SBMS_*` catalogues, `OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md`, `EVIDENCE_AND_VERIFICATION_REPORT.md`, this register and `FIGURE_PLACEHOLDER_CHECKLIST.md`. `CLAUDE_HANDOFF_GUIDE.md` §2 carries the full crosswalk.